

# GVIM 2.0: A Multi-Agent Research System for Chemistry and Materials Science

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## *Abstract*

Large language models are increasingly used as natural-language interfaces for chemistry and materials research, but their utility in scientific workflows depends on their ability to connect user requests with executable tools, structured data handling, and verifiable outputs. Here, we present GVIM 2.0, a multi-agent research system designed to support computational and informatics workflows in chemistry and materials science. GVIM 2.0 integrates language-model-based task planning with domain-specific Skills, package-backed computational tools, sandboxed execution, and artifact generation. On a 400-question public benchmark suite spanning ChemBench, MaScQA, and ChemLLMBench, the complete GVIM 2.0 workflow answered 325/400 questions correctly, achieving an accuracy of 81.25%, compared with 285/400 questions and 71.25% accuracy for a same-base API-only baseline. We further evaluated GVIM 2.0 on task-native workflows covering molecular solubility prediction, steel yield-strength prediction, reaction-table extraction, MS/MS spectral retrieval, chemical entity extraction, retrospective BACE inhibitor prioritization, Matbench experimental band-gap modelling, and temporal external validation of BACE1 activity prediction. These workflows generated inspectable outputs, including prediction files, metrics, figures, and reports. The results show that GVIM 2.0 can organize reproducible data-to-result workflows for selected chemistry and materials informatics tasks, providing a practical framework for tool-augmented scientific research assistance.



## **Toc**

### **1. Introduction**

Chemistry and materials science have traditionally relied on empirical judgement, heuristic rules, theoretical models, computational simulations, and repeated experimentation. Materials discovery and

chemical synthesis have been guided by manually curated databases, phase diagrams, design charts, expert intuition, and trial-and-error exploration [1-3]. In molecular design, reaction-condition optimization, and materials screening, researchers must search large chemical and compositional spaces for candidates with desired properties. This search is complicated by nonlinear relationships among molecular structure, synthetic route, processing condition, composition, microstructure, and final performance. These characteristics have motivated the increasing use of data-driven and algorithm-assisted methods in chemistry and materials research [1-4].

AI applications in chemistry and materials science predate the recent emergence of conversational LLMs. Expert systems, quantitative structure-activity relationship (QSAR) models, support vector machines, random forests, deep neural networks, graph neural networks, and machine-learning potentials have been used for molecular property prediction, drug discovery, reaction prediction, materials screening, and high-throughput computation [1,5]. Ramos et al. [5] reviewed the historical use of DENDRAL, QSAR, and later machine-learning approaches in cheminformatics and synthesis prediction. Mroz et al. [4] further described the expansion of AI from computation acceleration and chemical-space exploration to experimental planning, data-characterization pipelines, and laboratory automation. In this context, LLMs and agent systems represent a further stage in AI4Chem and AI4Mat rather than a replacement for established domain-specific models and computational methods.

The rapid development of large language models (LLMs) is closely associated with the Transformer architecture. The Transformer proposed by Vaswani et al. [6] is built upon the attention mechanism and overcomes the limitations of traditional recurrent neural networks and convolutional neural networks in sequence modeling, making it a key technical foundation for modern LLMs. After OpenAI released ChatGPT in November 2022, conversational LLMs entered public awareness through natural-language interaction, enabling researchers to perform tasks such as literature summarization, code generation, problem decomposition, preliminary experimental planning, and scientific writing assistance using ordinary language [7]. Pei et al. [8] noted that, since the release of ChatGPT, applications of NLP and LLMs in materials science have attracted rapidly increasing attention, with related studies covering literature mining, knowledge graphs, materials recommendation, automated materials discovery, and sustainable materials design. Unlike conventional machine-learning models, which typically rely on

structured data, manual feature engineering, and task-specific modeling, LLMs can process natural language, chemical language, semi-structured experimental records, and scientific literature, thereby providing a new technical basis for research assistance across tools, data sources, and tasks <sup>[5,8]</sup>.

Chemistry and materials science inherently rely on language to record, communicate, and reproduce experimental knowledge. Scientific papers, patents, laboratory records, reaction procedures, materials formulations, and safety protocols contain substantial tacit knowledge in textual form. Accordingly, NLP and LLMs are particularly suitable for extracting structured knowledge from scientific texts, including physical quantities, materials compositions, synthesis conditions, reaction pathways, named entities, performance metrics, and experimental observations <sup>[8–10]</sup>. Pei et al. <sup>[8]</sup> pointed out that NLP and language models can extract information from large volumes of text and support materials discovery, knowledge-graph construction, and sustainable materials design. Dagdelen et al. <sup>[9]</sup> demonstrated that fine-tuned LLMs can extract structured domain knowledge from complex scientific texts, thereby enabling the construction of larger-scale scientific databases. Polak et al. <sup>[10]</sup> further showed that general-purpose language models, combined with human supervision and prompt engineering, can extract materials data from full-text materials science papers and reduce the programming and domain-expertise barriers associated with data extraction. These studies indicate that the scientific value of LLMs lies not only in text generation, but also in transforming unstructured knowledge into searchable, computable, and reusable data resources.

Generative AI has also influenced molecular design and inverse problems in chemistry. Many chemical research tasks require identifying molecular structures, materials compositions, or synthetic routes from target properties. Conventional workflows rely on repeated design, synthesis or measurement, and evaluation cycles, whereas variational autoencoders, generative adversarial networks, reinforcement learning, Bayesian optimization, diffusion models, and LLM-based molecular generation provide computational strategies for accelerating this process [11]. Sumita et al. [11] reviewed progress in AI-based small-molecule design and emphasized continuing constraints related to training-data quality, synthetic accessibility, experimental validation, and measured performance improvement. Jablonka et al. [12] showed that fine-tuned LLMs can be applied to molecular and materials property prediction and reaction-yield prediction. Guo et al. [13]

evaluated LLMs on eight chemistry-related benchmark tasks and found competitive performance in some settings but clear limitations in more complex tasks, including reaction prediction. These results underline the need for task-level evaluation and domain-tool verification when applying LLMs to chemical problems.

For scientific use, LLM outputs need to be connected with external evidence, databases, and domain-specific tools. Model training data are temporally bounded, and generated responses may contain factual errors, incorrect citations, or unsupported reasoning. Retrieval-augmented generation (RAG) incorporates retrieved evidence from external documents or databases into generation, thereby improving factual grounding and source traceability. Lewis et al. [14] proposed the RAG framework, combining parametric memory with non-parametric retrieval for knowledge-intensive natural-language processing tasks. Gao et al. [15] reviewed RAG as an approach for improving factuality, updatability, and interpretability in LLM systems. Pei et al. [8] also discussed the role of RAG and knowledge graphs in improving LLM reliability in materials science.

Beyond RAG, tool use and agent technologies are further transforming LLMs from text-generation models into executors of scientific research workflows. Toolformer, proposed by Schick et al. [16], demonstrated that language models can learn when to invoke external APIs, how to pass parameters, and how to incorporate returned results into subsequent generation. The ReAct framework proposed by Yao et al. [17] alternates reasoning traces and action steps, enabling models to access external knowledge bases or environmental information during inference and dynamically update task plans. Wang et al. [18] conceptualized LLM-based autonomous agents as systems composed of perception, memory, planning, action, and tool-use modules, and noted that such agents are expanding from task execution in closed environments to complex scenarios in natural sciences and engineering. At the infrastructure level, interoperability protocols such as the Model Context Protocol (MCP) have begun to standardize connections between AI applications and external data sources or tool systems, providing an emerging engineering foundation for tool orchestration and context management in agent systems [19]. Together, these advances indicate that the integration of LLMs with RAG, tool use, memory mechanisms, and task-planning modules is accelerating the development of agentic workflows, enabling AI systems to progressively coordinate information retrieval, reasoning and analysis, tool invocation, and task execution.

In chemistry, LLM-based agent systems have demonstrated the value of combining language models with domain tools. ChemCrow integrates an LLM with expert-designed chemistry tools for tasks in organic synthesis, drug discovery, and materials design [20]. Coscientist demonstrates LLM-supported workflows for synthesis planning, public-data retrieval, hardware-documentation search, code execution, automated-equipment control, and experimental optimization [21]. These studies show that language models can function as interfaces and coordinators connecting literature, databases, computational models, experimental platforms, and human researchers. Related reviews also emphasize the importance of integrating LLMs with tools for synthetic-procedure search, property querying, robotic synthesis, and characterization workflows [5].

In materials science, AI is evolving along two parallel trajectories: task-specific AI and generalist AI. Task-specific AI is typically used for property prediction, structural characterization, experimental optimization, and materials screening, whereas generalist AI places greater emphasis on knowledge representation, reasoning and planning, tool orchestration, and human–AI collaboration. Li et al. [2] described AI4Mat as two complementary pathways: task-specific AI functions as a “solver” for specific scientific problems, while generalist AI acts as a “coordinator” that connects knowledge, tools, and workflows; together, these pathways point toward automated materials discovery and self-driving laboratories. This framework suggests that future intelligent systems for chemistry and materials science should not simply replace domain-specific models with a single general model, but should instead integrate domain models, knowledge bases, experimental data, simulation tools, and agentic workflows<sup>[2,8]</sup>. Pei et al. [8] also emphasized that the potential of language models in materials research lies not only in text processing, but also in their integration with knowledge graphs, RAG, automated experimentation, and sustainable materials design.

Automated experimentation and self-driving laboratories represent important directions in AI for Chemistry and AI for Materials. Self-driving laboratories generally integrate robotic experimental platforms, online characterization systems, data-analysis software, and decision-making algorithms, enabling closed-loop experimental design, execution, real-time analysis, and parameter optimization. Tom et al. [22] noted that self-driving laboratories can significantly

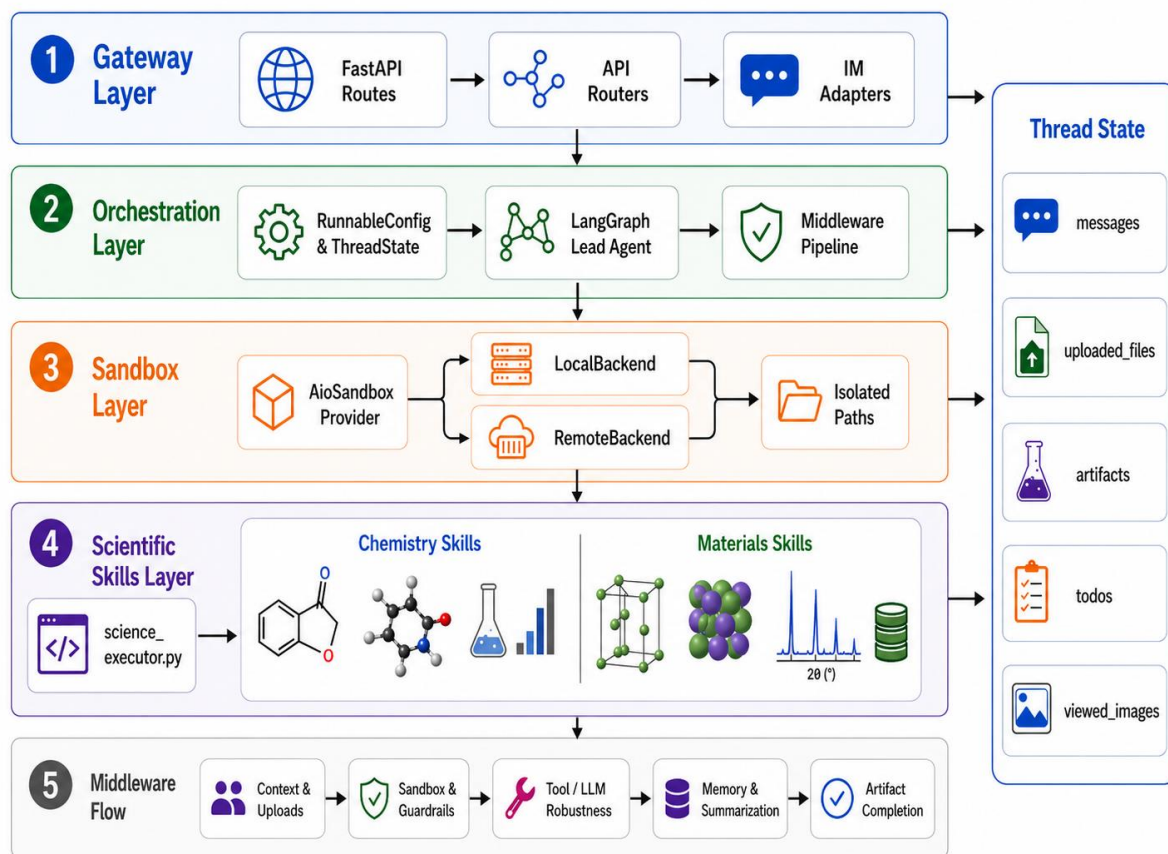
accelerate chemistry and materials discovery through experimental-process automation and autonomous experimental planning. Häse et al. [23] discussed the role of next-generation self-driving laboratories in experimental optimization and automated discovery; Burger et al. [24] demonstrated a mobile robotic chemist for autonomous search of photocatalytic systems; Kusne et al. [25] showed Bayesian active learning in closed-loop materials discovery; Raccuglia et al. [26] demonstrated that failed experimental data can also support machine-learning-assisted materials discovery; Granda et al. [27] used machine learning to control an organic-synthesis robot in the search for new reactivity; MacLeod et al. [28] constructed a self-driving laboratory for accelerated thin-film materials discovery; and Shields et al. [29] demonstrated the value of Bayesian optimization in chemical reaction optimization. Together, these studies indicate that the integration of AI with automated experimental platforms is driving chemistry and materials research from offline prediction toward online decision-making and closed-loop validation.

The deployment of LLMs, agents, and automated experimentation in scientific workflows still faces technical and scientific challenges. LLMs may produce hallucinated statements, incorrect citations, unstable reasoning, or irreproducible outputs, which are especially problematic when chemical reactions, materials formulations, or safety-related decisions are involved [5,8,13,30]. Chemical and materials datasets are also heterogeneous: experimental conditions, failed experiments, negative results, and processing parameters are often incompletely recorded, leading to bias in model training and validation [1,3,26]. Reviews in chemistry and materials informatics emphasize continuing issues related to data quality, model interpretability, benchmark standardization, experimental validation, and extrapolation to new chemical or materials spaces [3,5].

These developments motivate research systems that combine natural-language interaction with domain tools, executable workflows, and traceable outputs. In this work, we build upon the DeerFlow 2.0 framework [41] to develop GVIM 2.0, a multi-agent system for chemistry and materials science. GVIM 2.0 uses domain-specific Skills [42] as reusable modules for computational chemistry, materials analysis, machine-learning modelling, scientific visualization, literature processing, and artifact generation. The system is evaluated using public chemistry and materials benchmarks and task-native workflows with objective output metrics. The present study therefore examines whether an agentic workflow can convert user-facing scientific tasks into executable procedures and inspectable research artifacts.

## 2. System Architecture and Evaluation Design

Built upon the DeerFlow 2.0 framework [41], GVIM 2.0 adapts a general-purpose workflow-agent architecture for chemistry and materials science. A typical run begins at the web front end, where the user provides a scientific task, uploads data files, and requests output artifacts. The gateway layer standardizes these inputs and forwards them to the orchestration layer. The lead agent then parses the task, decomposes it into executable steps, and selects the required domain Skills or computational tools [20-25,32,33]. GVIM 2.0 separates language-model-based planning from package-backed scientific execution. The language model is used for task planning, parameter routing, and workflow coordination, whereas molecular structures, crystal properties, diffraction patterns, numerical descriptors, and prediction files are generated by external tools and domain Skills. Chemistry-related Skills support molecular sketching, RDKit descriptor calculation, 3D conformer generation, reaction checking, scaffold or fragment extraction, and molecular similarity analysis. Materials-related Skills support structure parsing, space-group and density analysis, local coordination analysis, X-ray diffraction simulation, precursor stoichiometry calculation, and database retrieval through materials data interfaces. The Skill-system design also allows additional chemistry- and materials-oriented modules to be integrated without changing the core orchestration framework. Executable operations are performed in an isolated sandbox environment, where uploaded files, intermediate results, scripts, and final outputs are maintained within thread-specific workspaces. GVIM 2.0 uses a middleware-controlled artifact completion mechanism to check whether requested outputs have been generated before the final response is returned. Requested artifacts, including molecular drawings, 3D conformers, materials charts, prediction files, and reports, are matched against tool-execution records in the thread workspace. When an expected artifact is absent, the workflow is redirected to complete the corresponding execution step. This design records the input files, execution steps, intermediate outputs, and final artifacts associated with each workflow.



**Figure 1. GVIM 2.0 system architecture**

The evaluation design combines benchmark-based assessment with workflow-level evaluation. Public benchmarks were used to measure domain question-answering performance across chemistry- and materials-related tasks. In parallel, task-native workflows were used to evaluate whether GVIM 2.0 could coordinate multi-step procedures and generate outputs that could be independently scored. Depending on the task type, performance was quantified using prediction error, extraction precision/recall/F1, retrieval top-k accuracy, mean reciprocal rank, enrichment factor, and recall under fixed-budget prioritization.

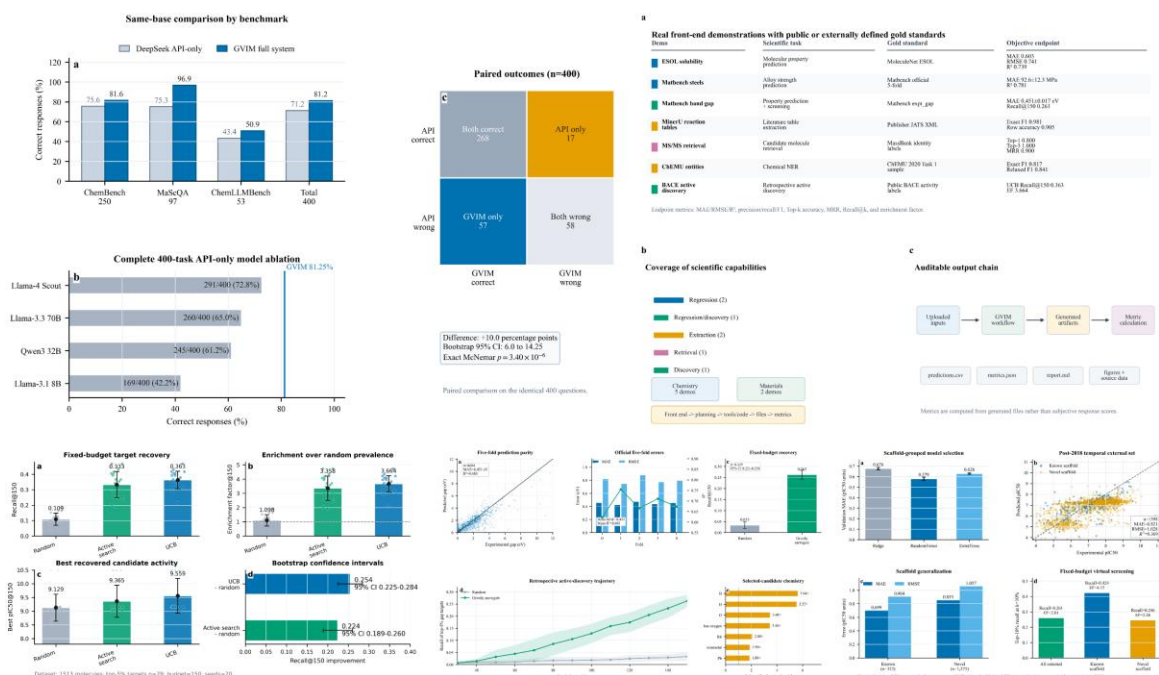
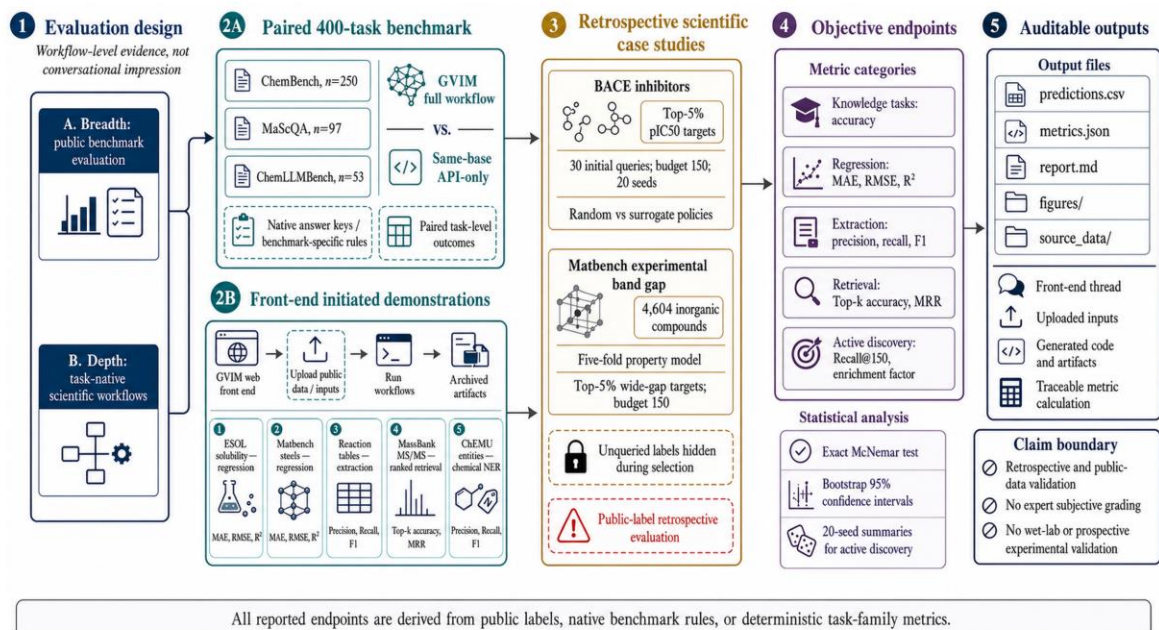


Figure 2. Experimental design and evaluation workflow of GVIM 2.0

## 3. Methods

### 3.1 Benchmark construction and scoring

The benchmark suite contained 400 public chemistry and materials questions: 250 ChemBench questions, 97 MaScQA questions, and 53 ChemLLMBench questions. GVIM outputs and API-only outputs were scored with the native answer key or benchmark-specific scoring rule for each task family. Because the three sources use different scoring definitions,

the 400-question total is reported as a descriptive aggregate, while per-benchmark results remain the primary interpretable units.

### 3.2 Task-native workflow scoring

Regression workflows used MAE, RMSE, and  $R^2$  on held-out labeled data. Extraction workflows used exact or relaxed precision, recall, and F1 after deterministic normalization where appropriate. Retrieval workflows used top-k accuracy and mean reciprocal rank. These metrics follow standard definitions for the corresponding task families.

### 3.3 Statistical analysis

For paired benchmark comparison, we used exact McNemar testing on the full-vs-API contingency table and bootstrap confidence intervals for the paired accuracy difference. For retrospective prioritization workflows, endpoint differences between surrogate-guided selection policies and random selection were summarized across random seeds and evaluated with bootstrap confidence intervals.

### 3.4 BACE1 temporal external-validation protocol

Public ChEMBL BACE1 IC50 records were standardized to pIC50 and aggregated by compound. Training compounds were restricted to document years  $\leq 2015$ ; 2016–2017 records were excluded; and external compounds were restricted to document years  $\geq 2018$ . Exact molecule overlap was removed by InChIKey. Model selection used only the historical training file and three-fold GroupKFold grouped by Bemis–Murcko scaffold. Candidate models were Ridge, RandomForestRegressor and ExtraTreesRegressor using Morgan fingerprints (radius 2, 1,024 bits) plus eight RDKit descriptors. The prespecified selection criterion was mean validation MAE.

The selected model was refit to all historical compounds and generated external\_predictions.csv before gold labels were supplied. Stage 2 verified row alignment and the frozen prediction content, joined labels one-to-one by compound\_id, and did not retrain or tune a model. MAE, RMSE and  $R^2$  used their standard definitions. For ranking evaluation, actives were the highest external pIC50 decile and the query set was the highest predicted-pIC50 decile, with deterministic compound\_id tie breaking. Recall@10% was the fraction of top-decile actives recovered; EF10 divided the selected-set active fraction by external active

prevalence. Metrics were recomputed separately for all, known-scaffold and scaffold-novel compounds.

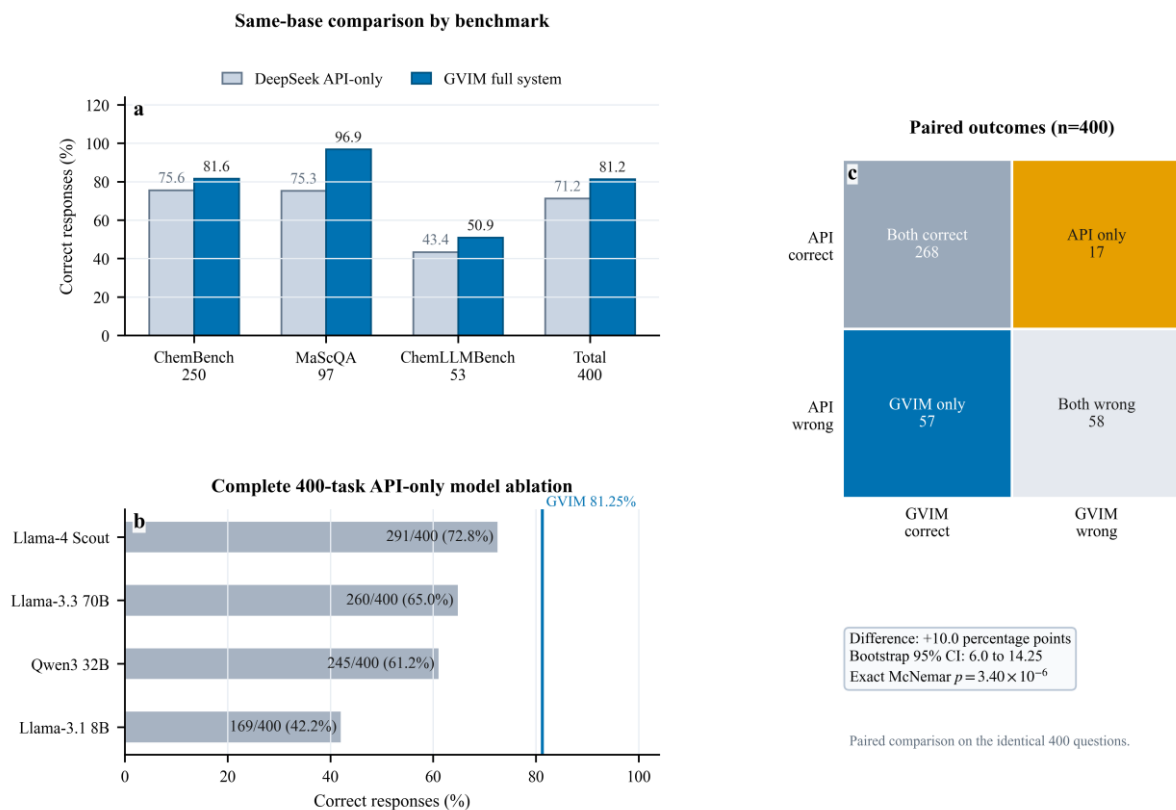
## 4. Benchmark Evaluation on Public Chemistry and Materials Tasks

We evaluated GVIM 2.0 on 400 public chemistry and materials questions assembled from ChemBench, MaScQA, and ChemLLMBench [13,34,35]. The suite combines the previously completed 360-question evaluation with an additional 40-question public extension. All questions were submitted through the GVIM workflow, and scoring followed the native answer key or task-specific rule for each benchmark. The complete GVIM 2.0 workflow was compared with a same-base API-only baseline, in which the underlying model was used without the GVIM workflow layer.

**Table 1. Public benchmark performance of GVIM 2.0 and the same-base API-only baseline.**

| Benchmark         | N   | GVIM correct | GVIM (%) | API-only correct | API-only (%) |
|-------------------|-----|--------------|----------|------------------|--------------|
| ChemBench         | 250 | 204/250      | 81.60    | 189/250          | 75.60        |
| MaScQA            | 97  | 94/97        | 96.91    | 73/97            | 75.26        |
| ChemLLMBench      | 53  | 27/53        | 50.94    | 23/53            | 43.40        |
| Descriptive total | 400 | 325/400      | 81.25    | 285/400          | 71.25        |

Across all 400 questions, GVIM 2.0 increased the descriptive total accuracy from 71.25% to 81.25%. The paired analysis showed 57 cases solved only by GVIM 2.0 and 17 cases solved only by the API-only baseline. This corresponds to a paired improvement of 10.00 percentage points, with a McNemar exact two-sided p value of  $3.40 \times 10^{-6}$  and a bootstrap 95% confidence interval of 6.00-14.25 percentage points. The complete GVIM 2.0 workflow therefore achieved higher paired accuracy than the same-base API-only baseline on this benchmark suite.



**Figure 3. Public benchmark performance of GVIM. (a)** Correct-response accuracy for ChemBench, MaScQA, ChemLLMBench, and the descriptive 400-question total for the same-base DeepSeek API-only setting and the full GVIM workflow; numbers above bars are percentages. **(b)** Accuracy of four additional API-only reference models on the identical 400-question suite; the vertical line marks GVIM full-system accuracy (81.25%). **(c)** Paired outcome matrix for GVIM versus the same-base API-only setting. The paired difference was 10.0 percentage points (bootstrap 95% CI, 6.00–14.25 percentage points; exact two-sided McNemar  $p = 3.40 \times 10^{-6}$ ).

## 4.1 Front-End Workflow Evaluations

We next evaluated front-end workflows spanning molecular prediction, alloy-property prediction, literature table extraction, spectral retrieval, and chemical entity extraction. For ESOL and Matbench steels fold 0, the front end generated frozen predictions from unlabeled test inputs; held-out metrics were calculated only afterward by joining those predictions to withheld public labels. The other demonstrations were scored from archived front-end artifacts using public or publisher-defined references and standard task-family metrics: MAE, RMSE, and  $R^2$  for regression; precision, recall, and F1 for extraction; and top-k accuracy and mean reciprocal rank for retrieval.

Table 2. Verified front-end demonstrations and objective endpoint metrics. Held-out ESOL and Matbench steels fold-0 metrics were computed post hoc from frozen front-end predictions; they were not available to

the agent during execution.

| Evaluation workflow       | Reference standard                                                   | Objective endpoint                                       | Scale     |
|---------------------------|----------------------------------------------------------------------|----------------------------------------------------------|-----------|
| ESOL solubility           | MoleculeNet ESOL                                                     | MAE = 0.603; RMSE = 0.741; $R^2$ = 0.739                 | n = 254   |
| Matbench steels fold 0    | Matbench v0.1 official fold 0;<br>frozen predictions scored post hoc | MAE = 111.444 MPa; RMSE = 164.320 MPa; $R^2$ = 0.684     | n = 63    |
| Reaction table extraction | Publisher JATS XML                                                   | Exact P/R/F1 = 0.981/0.981/0.981; row accuracy = 0.905   | 105 cells |
| MS/MS retrieval           | MassBank InChIKey                                                    | Top-1 = 0.800; Top-3 = 1.000; Top-5 = 1.000; MRR = 0.900 | 5 queries |
| ChEMU entity extraction   | ChEMU Task 1 sample                                                  | Exact P/R/F1 = 0.931/0.728/0.817; relaxed F1 = 0.841     | 92 spans  |

The verified workflows show complementary aspects of the system. ESOL and Matbench steels fold 0 assess lightweight molecular and materials property-prediction workflows [36-38]. The steels thread is a single-fold front-end demonstration: official fold 0 was uploaded, while a shuffled five-fold split of its 249-row training partition was used internally for model selection. Reaction-table extraction uses publisher XML as the gold standard, MS/MS retrieval uses public MassBank InChIKey identities, and ChEMU uses exact and relaxed span scoring [39,40].

a

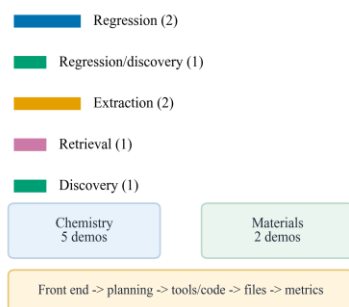
## Real front-end demonstrations with public or externally defined gold standards

| Demo                   | Scientific task                 | Gold standard                           | Objective endpoint                                |
|------------------------|---------------------------------|-----------------------------------------|---------------------------------------------------|
| ESOL solubility        | Molecular property prediction   | MoleculeNet ESOL                        | MAE 0.603<br>RMSE 0.741<br>R <sup>2</sup> 0.739   |
| Matbench steels fold 0 | Alloy strength prediction       | Official fold 0; withheld-label scoring | Test MAE 111.444 MPa<br>Test R <sup>2</sup> 0.684 |
| Matbench band gap      | Property prediction + screening | Matbench expt_gap                       | MAE 0.451±0.017 eV<br>Recall@150 0.263            |
| MinerU reaction tables | Literature table extraction     | Publisher JATS XML                      | Exact F1 0.981<br>Row accuracy 0.905              |
| MS/MS retrieval        | Candidate molecule retrieval    | MassBank identity labels                | Top-1 0.800<br>Top-3 1.000<br>MRR 0.900           |
| ChEMU entities         | Chemical NER                    | ChEMU 2020 Task 1 sample                | Exact F1 0.817<br>Relaxed F1 0.841                |
| BACE active discovery  | Retrospective active discovery  | Public BACE activity labels             | UCB Recall@150 0.363<br>EF 3.664                  |

Endpoint metrics: MAE/RMSE/R<sup>2</sup>, precision/recall/F1, Top-k accuracy, MRR, Recall@k, and enrichment factor.

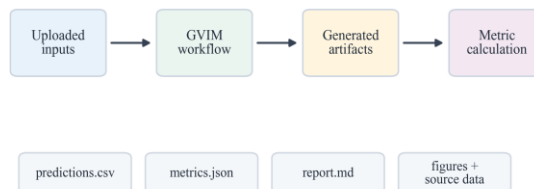
b

## Coverage of scientific capabilities



c

## Auditable output chain



Metrics are computed from generated files rather than subjective response scores.

Figure 4. Task-native chemistry and materials demonstrations initiated through the GVIM front end. (a) MoleculeNet ESOL: frozen predictions for 254 unlabeled test molecules were scored afterward against withheld public labels (MAE = 0.603, RMSE = 0.741,  $R^2$  = 0.739). (b) Matbench steels fold 0: frozen predictions for 63 unlabeled test alloys were scored afterward against withheld labels (MAE = 111.444 MPa, RMSE = 164.320 MPa,  $R^2$  = 0.684). This is a single-fold front-end demonstration, not the full Matbench five-fold protocol. (c) Publisher-JATS reaction-table extraction over 105 cells. (d) MassBank candidate retrieval over five queries. (e) ChEMU entity extraction over 92 gold spans. (f) Coverage of the five workflow types.

## 4.2 Retrospective Prioritization of BACE Inhibitor Candidates

We evaluated GVIM 2.0 on a retrospective fixed-budget prioritization task using a public BACE molecular property dataset [36]. The dataset was treated as a closed candidate library with pIC50 as the activity label. High-activity targets were defined as the top 5% of molecules by pIC50. Starting from 30 randomly queried molecules, each policy selected batches of 10 candidates up to a total budget of 150 queries over 20 random seeds. During each simulated run, model training and candidate selection used only labels that had already been queried; the remaining pIC50 values were used only as the oracle when a molecule was selected and for final metric computation.

Table 3. Retrospective BACE prioritization performance under a fixed query budget.

| Policy              | Recall@150 | Hit rate@150 | EF@150 | Best pIC50@150 |
|---------------------|------------|--------------|--------|----------------|
| Random              | 0.109      | 0.057        | 1.098  | 9.129          |
| Greedy-surrogate ET | 0.333      | 0.175        | 3.358  | 9.365          |
| UCB-surrogate ET    | 0.363      | 0.191        | 3.664  | 9.559          |

In the BACE prioritization workflow, the UCB ExtraTrees surrogate policy increased Recall@150 from 0.109 for random selection to 0.363, corresponding to an enrichment factor of 3.664. The bootstrap 95% confidence interval for UCB minus random Recall@150 was 0.225-0.284, using 10,000 paired per-seed bootstrap resamples. The workflow produced the required JSON, CSV, PNG, and Markdown artifacts.

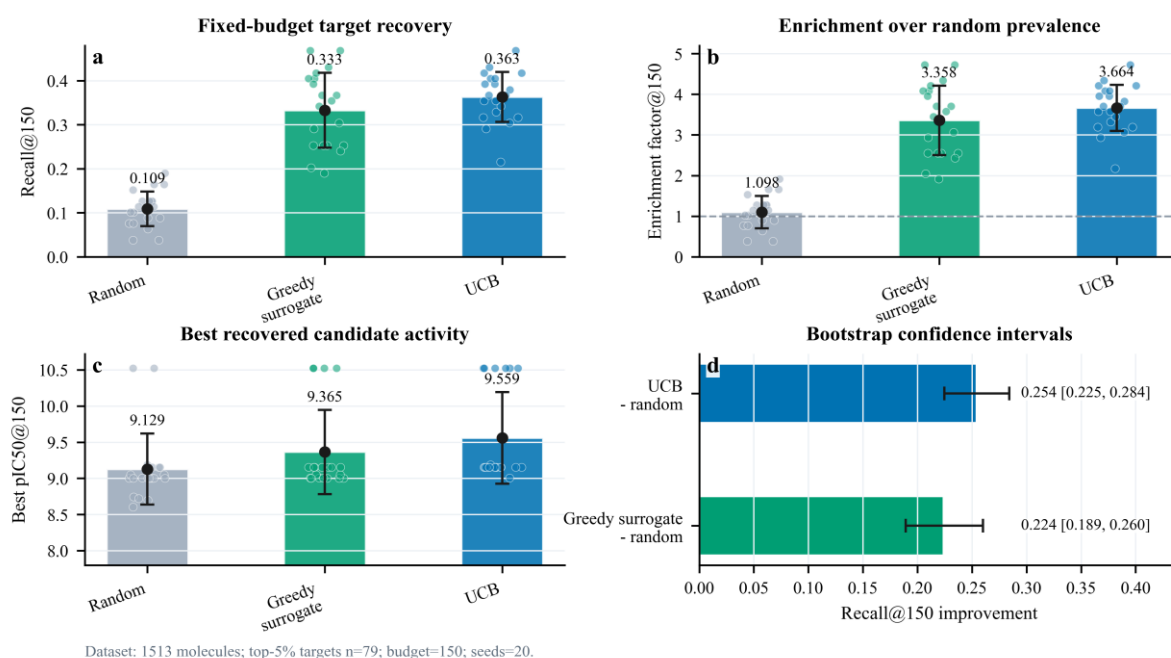


Figure 5. Retrospective BACE prioritization under a fixed query budget using public pIC50 labels. High-activity targets were the top 5% of 1,513 molecules ( $n = 79$ ). Each run began with 30 random queries and selected batches of 10 up to 150 queries using an ExtraTrees surrogate; 20 random seeds were used. (a) Recall@150, (b) enrichment factor@150, and (c) best recovered pIC50 for random, greedy-surrogate and upper-confidence-bound policies. Bars show means, points show individual seeds, and error bars show  $\pm 1$  SD; the dashed line in (b) denotes random-prevalence enrichment ( $EF = 1$ ). (d) Mean recall improvement over random with 95% confidence intervals from 10,000 paired per-seed bootstrap resamples.

### 4.3 Summary of Archived GVIM Workflow Outputs

Table 4 summarizes the chemistry and materials evaluation workflows supported by archived GVIM

output artifacts.

Table 4. Archived GVIM workflow outputs and objective metrics.

| Evaluation workflow                | Reference dataset or standard                                                              | Artifact-derived objective metrics                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| ESOL solubility prediction         | MoleculeNet ESOL                                                                           | MAE = 0.603; RMSE = 0.741; $R^2$ = 0.739; n = 254                                                                     |
| Matbench steels fold 0             | Matbench v0.1 official fold 0; front-end predictions with post-hoc withheld-label scoring  | MAE = 111.444 MPa; RMSE = 164.320 MPa; $R^2$ = 0.684; n = 63                                                          |
| MinerU reaction-table extraction   | Publisher JATS XML table nanomaterials-12-01070-t002                                       | Exact precision = 0.981; recall = 0.981; F1 = 0.981; row accuracy = 0.905                                             |
| MS/MS candidate retrieval          | MassBank InChIKey identity from public MassBank-data records                               | Top-1 = 0.800; Top-3 = 1.000; Top-5 = 1.000; MRR = 0.900; n_queries = 5                                               |
| ChEMU entity extraction            | ChEMU 2020 Task 1 sample v3                                                                | Exact P/R/F1 = 0.931/0.728/0.817; relaxed F1 = 0.841; gold spans = 92                                                 |
| BACE prioritization                | bace_active_discovery.csv                                                                  | Random Recall@150 = 0.109; UCB Recall@150 = 0.363; EF@150 = 3.664; 95% CI = 0.225–0.284                               |
| Matbench band-gap main case        | Matbench v0.1 matbench_expt_gap                                                            | MAE = $0.4513 \pm 0.0170$ eV; RMSE = $0.8049 \pm 0.0456$ eV; $R^2$ = $0.6854 \pm 0.0445$ ; greedy Recall@150 = 0.2626 |
| BACE1 temporal external validation | ChEMBL ChEMBL4822; $\leq 2015$ training and $\geq 2018$ non-overlapping external compounds | MAE=0.8215, RMSE=1.0278, $R^2$ =0.3687; Recall@10%=0.2609; EF10=2.6054; n_external=1,598                              |

## 4.4 Retrospective Prioritization of High-Band-Gap Inorganic Compounds

A materials-focused workflow was evaluated using the public Matbench v0.1 matbench\_expt\_gap dataset as a closed retrospective library. GVIM 2.0 organized both property modelling and fixed-budget prioritization from uploaded input files. The archived workflow outputs were used for regression evaluation, candidate prioritization, and element-enrichment analysis.

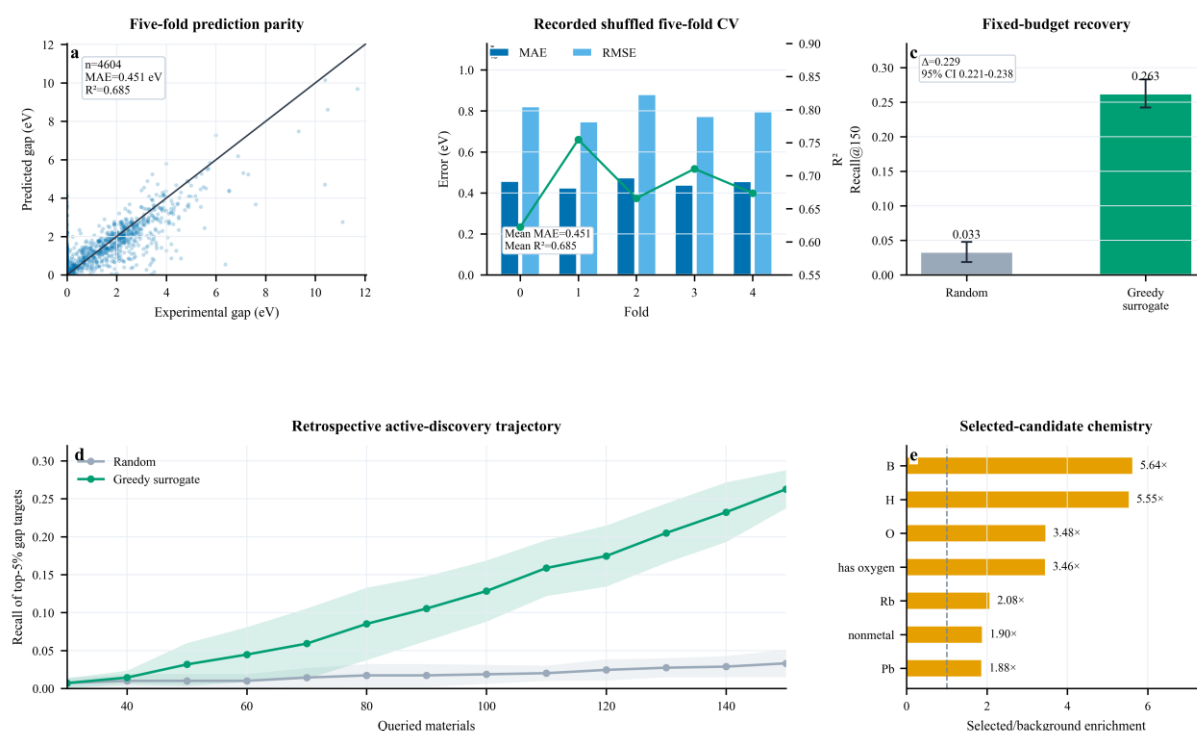


Figure 6. Retrospective Matbench experimental band-gap prioritization generated from archived GVIM front-end artifacts. (a) Out-of-fold prediction parity for 4,604 experimentally measured inorganic compounds using the recorded composition-only model. (b) Per-fold MAE, RMSE, and  $R^2$  under KFold( $n\_splits = 5$ ,  $shuffle = True$ ,  $random\_state = 18012019$ ); this is the workflow's recorded shuffled KFold rule, not an official Matbench split. (c) Recall@150 for random and greedy-surrogate policies after 30 initial random queries and batches of 10 to a total budget of 150; bars show means over seeds 0, 1, and 2, and error bars show  $\pm 1$  SD. The inset reports the archived 95% interval for the mean paired recall difference; with only three seeds, it is interpreted descriptively. (d) Mean recall trajectories across the same three seeds; shaded bands show  $\pm 1$  SD. (e) Selected-to-background enrichment of the seven most enriched recorded composition features; the dashed line marks an enrichment ratio of 1.

Table 5. Matbench experimental band-gap prioritization artifact summary.

| Item                    | Value from GVIM artifact                                                                                                       |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Dataset                 | Matbench v0.1 matbench_expt_gap; $n = 4604$ compounds; target = gap_expt_eV                                                    |
| Feature set             | 98 precomputed composition descriptors from the uploaded fast-feature table                                                    |
| Model and CV rule       | HistGradientBoostingRegressor; KFold( $n\_splits = 5$ , $shuffle = True$ , $random\_state = 18012019$ )                        |
| Regression metrics      | MAE = $0.4513 \pm 0.0170$ eV; RMSE = $0.8049 \pm 0.0456$ eV; $R^2 = 0.6854 \pm 0.0445$                                         |
| Prioritization target   | Top 5% highest-gap materials; threshold = 3.7385 eV; $n\_top5 = 231$                                                           |
| Prioritization settings | Seeds = [0, 1, 2]; initial_random_size = 30; batch_size = 10; total query budget = 150; policies = random and greedy_surrogate |

| Item                   | Value from GVIM artifact                                                                                                                                                                                                                                           |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recall@150             | Random = $0.0332 \pm 0.0180$ ; greedy_surrogate = $0.2626 \pm 0.0250$ ; difference = 0.2294 with 95% CI 0.2208–0.2381                                                                                                                                              |
| Hit rate and best gap  | Hit rate@150: random = 0.0511, greedy = 0.4044; best gap@150: random = 7.013 eV, greedy = 9.033 eV                                                                                                                                                                 |
| Runtime and provenance | runtime = 73.8 s; data_hash = 178a3b4a4009                                                                                                                                                                                                                         |
| Output files           | metrics.json; predictions_5fold.csv; active_discovery_trajectories.csv; selected_candidates.csv; element_enrichment.csv; source_data_regression.csv; source_data_active_discovery.csv; figure_regression_parity.png; figure_active_discovery_recall.png; report.md |

The archived GVIM output shows that the lightweight composition-only model achieved a shuffled five-fold KFold MAE of 0.4513 eV and  $R^2$  of 0.6854 on all 4,604 records. This workflow is distinct from the Matbench steels fold-0 case. In the retrospective prioritization simulation, the greedy-surrogate policy increased Recall@150 from 0.0332 to 0.2626. The archived output reports a 95% interval of 0.2208-0.2381 for the mean paired recall difference.

**Table 6. Per-fold regression performance for the Matbench experimental band-gap case.**

| Fold | MAE (eV) | RMSE (eV) | $R^2$  |
|------|----------|-----------|--------|
| 0    | 0.4585   | 0.8223    | 0.6224 |
| 1    | 0.4261   | 0.7488    | 0.7547 |
| 2    | 0.4756   | 0.8819    | 0.6659 |
| 3    | 0.4397   | 0.7741    | 0.7105 |
| 4    | 0.4568   | 0.7974    | 0.6736 |

Element-enrichment analysis in the generated report identified the strongest selected-versus-background enrichment for frac\_B (5.64x), frac\_H (5.55x), frac\_O (3.48x), has\_oxygen (3.46x), and frac\_Rb (2.08x).

## 4.5 Temporal External Validation of BACE1 Activity

Temporal generalization was evaluated using a two-stage front-end workflow for human BACE1 (ChEMBL target ChEMBL4822). Stage 1 used 5,296 compounds associated with documents published through 2015 for all feature design and model selection. Measurements from 2016-2017 were excluded as a temporal buffer, and 1,598 non-overlapping compounds associated with documents from 2018 onward formed the external set. Exact molecule overlap was removed by InChIKey; 1,275 external compounds

(79.8%) were scaffold-novel relative to training. External pIC50 labels were not provided during Stage 1. Model selection used three-fold GroupKFold grouped by Murcko scaffold with Morgan fingerprints (radius 2, 1,024 bits) and eight RDKit descriptors. RandomForestRegressor was selected by the lowest mean validation MAE ( $0.579 \pm 0.016$ ), compared with Ridge (0.676) and ExtraTrees (0.628). After predictions were frozen, Stage 2 joined the public external labels by compound\_id without retraining. The selected model achieved MAE = 0.822, RMSE = 1.028, and  $R^2 = 0.369$  on 1,598 external compounds. This performance was higher than that of Ridge (MAE = 0.953,  $R^2 = 0.105$ ) and the mean baseline (MAE = 1.185,  $R^2 = -0.123$ ). At a 10% screening budget, the model recovered 42 of 161 top-decile external actives (Recall@10% = 0.261; EF10 = 2.61). Known-scaffold compounds gave Recall@10% = 0.424 and EF10 = 4.15, whereas the scaffold-novel subset retained non-random prioritization (Recall@10% = 0.246; EF10 = 2.46) but showed higher prediction error.

Table 7. Prespecified BACE1 temporal external-validation endpoints derived from the frozen GVIM front-end artifacts.

| Endpoint            | Model/subset   | Value                              | Scope     |
|---------------------|----------------|------------------------------------|-----------|
| Scaffold CV MAE     | RandomForest   | $0.579 \pm 0.016$                  | 3 folds   |
| External regression | RandomForest   | MAE 0.822; RMSE 1.028; $R^2$ 0.369 | n = 1,598 |
| External regression | Ridge          | MAE 0.953; RMSE 1.224; $R^2$ 0.105 | n = 1,598 |
| Top-10% screening   | All external   | Recall 0.261; EF 2.61              | k = 160   |
| Top-10% screening   | Known scaffold | Recall 0.424; EF 4.15              | n = 323   |
| Top-10% screening   | Novel scaffold | Recall 0.246; EF 2.46              | n = 1,275 |

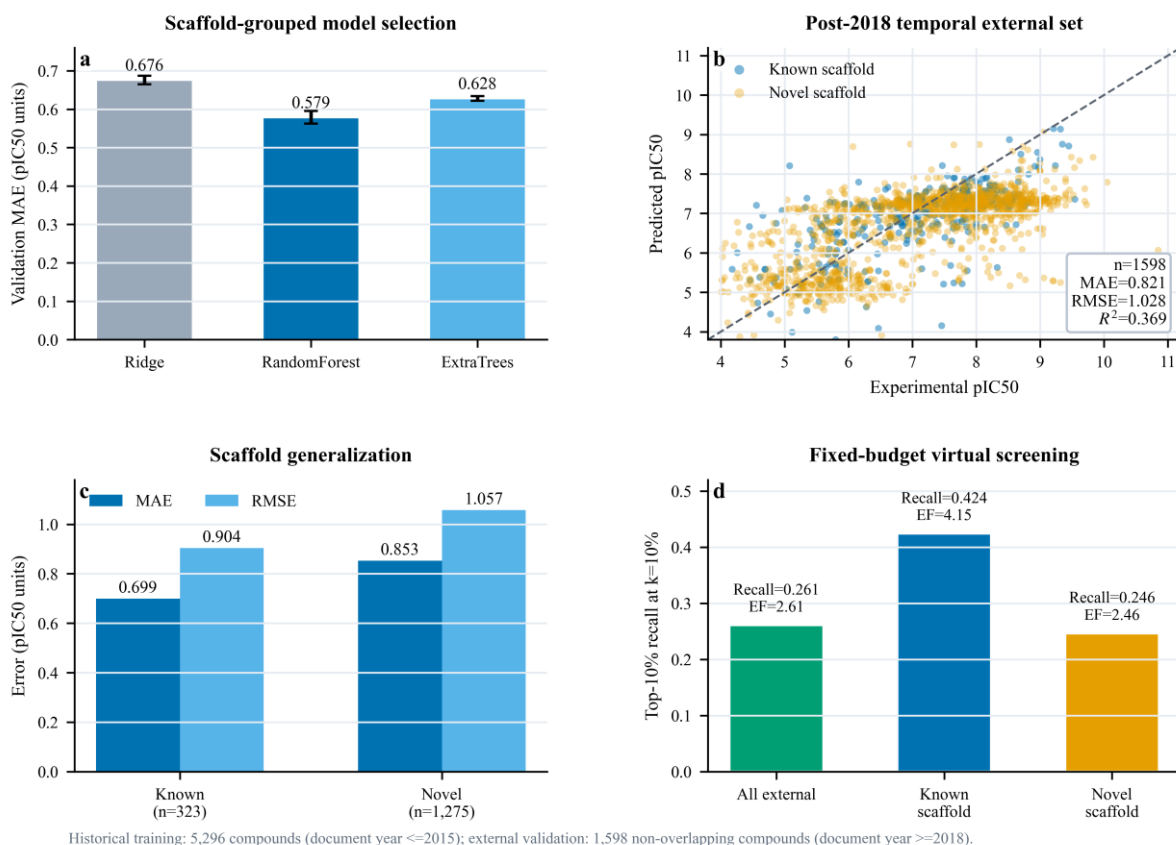


Figure 7. Label-withheld temporal external evaluation of BACE1 activity using public ChEMBL labels. Historical compounds associated with documents through 2015 were used for model selection, 2016–2017 measurements were excluded, and non-overlapping compounds from 2018 onward formed the external set. (a) Three-fold Murcko-scaffold GroupKFold MAE for the prespecified candidate models; bars show fold means and error bars show  $\pm 1$  SD. (b) Frozen RandomForest predictions versus external pIC50 for known and novel scaffolds; the dashed line is  $y = x$ . (c) MAE and RMSE for known- and novel-scaffold subsets. (d) Recall of top-decile actives at a 10% selection budget; labels report the corresponding enrichment factor.

## 5. Discussion

GVIM 2.0 was implemented as an end-to-end research-assistance system for chemistry and materials science. The system connects natural-language scientific requests with executable computational procedures, domain-specific tools, and auditable research artifacts. Rather than functioning as a stand-alone question-answering model, GVIM 2.0 provides a workflow environment in which scientific tasks can be interpreted, executed, documented, and evaluated through objective outputs. The paired 400-question benchmark provides an initial quantitative evaluation of system performance. Across chemistry- and materials-related questions, the complete GVIM 2.0 workflow answered 325 out of 400 questions correctly,

corresponding to an accuracy of 81.25%. Under the same evaluation setting, the API-only baseline using the same underlying model answered 285 questions correctly, corresponding to an accuracy of 71.25%. The 10.0 percentage-point improvement indicates that the GVIM 2.0 workflow can improve benchmark performance by structuring model reasoning, domain-knowledge access, tool use, and answer generation. This result supports the value of workflow-level design for improving the reliability of large-language-model-based scientific assistance.

Beyond benchmark question answering, GVIM 2.0 was evaluated through a series of task-native chemistry and materials workflows. These tasks covered molecular solubility prediction, alloy yield-strength prediction, reaction-table extraction, MS/MS candidate retrieval, chemical entity extraction, retrospective BACE candidate prioritization, materials property modelling, and temporal external validation of BACE1 activity. In each task, the system generated inspectable research artifacts, including prediction tables, metric summaries, figures, intermediate files, and final reports. These outputs make the workflows traceable and allow the reported results to be evaluated using public labels, benchmark rules, publisher XML, spectral identities, or established retrieval metrics. The molecular and materials prediction tasks demonstrate the ability of GVIM 2.0 to construct machine-learning workflows from user-facing scientific inputs. For molecular solubility and alloy yield-strength prediction, the system organized data loading, feature generation, model training, validation, prediction, and report generation into complete computational pipelines. The resulting files contain not only final predictions but also metadata and evaluation summaries, allowing the modelling process to be inspected after execution. These examples show that GVIM 2.0 can translate common cheminformatics and materials-informatics requests into executable workflows that produce reusable outputs. The extraction and retrieval tasks further demonstrate the task coverage of the system. In reaction-table extraction, GVIM 2.0 processed publisher XML and generated structured reaction information from literature-derived inputs. In the ChEMU chemical entity extraction task, the system produced entity-level predictions that could be evaluated using extraction metrics. In the MS/MS candidate retrieval workflow, the system ranked candidate spectra according to established retrieval protocols and reported top-k performance. Together, these examples show that GVIM 2.0 can operate across heterogeneous task formats, including structured documents, chemical text, molecular identifiers, and spectral-retrieval data. The

retrospective BACE prioritization workflow illustrates the ability of the system to organize model-guided virtual screening under a fixed-budget setting. Using a UCB ExtraTrees surrogate model, GVIM 2.0 prioritized compounds from an existing candidate library and recovered more high-activity compounds than random selection. The enrichment over the background prevalence of active compounds indicates that the system can execute an active-search-style workflow from input data to final prioritization metrics. Since both the candidate library and activity labels were derived from a public dataset, this case is best interpreted as a retrospective prioritization study. Nevertheless, it provides a concrete example of how GVIM 2.0 can support molecular screening workflows in a reproducible computational setting. The Matbench experimental band-gap case provides a materials-focused workflow demonstration. GVIM 2.0 generated a complete composition-based modelling pipeline for inorganic compounds and produced regression outputs, active-prioritization trajectories, selected candidates, feature-enrichment results, and visualization files. These artifacts show that the system can organize uploaded data into a materials-informatics analysis workflow and generate structured outputs. This workflow also demonstrates how natural-language requests can be converted into a sequence of executable modelling steps, including data preprocessing, model fitting, candidate ranking, and figure generation. The temporal BACE1 validation experiment provides a stricter test of external generalization. Historical compounds were used for model selection, while external labels were withheld until after the prediction files had been finalized. Final evaluation was then performed on non-overlapping post-2018 compounds. The selected model outperformed both a linear baseline and a mean predictor and recovered top-decile external actives above random expectation. Performance decreased for scaffold-novel compounds, reflecting the known difficulty of extrapolating molecular activity models to new chemical scaffolds. This result illustrates both the utility and the boundary of the workflow: GVIM 2.0 can organize data-isolated prediction and evaluation, while model performance remains governed by the available data, molecular representation, model class, and validation design. Taken together, these evaluations show that GVIM 2.0 can support diverse chemistry and materials workflows through a unified research-assistance interface. The system does not merely generate conversational answers; it creates executable procedures, saves intermediate and final artifacts, and reports objective metrics. This artifact-centered design is particularly

important for scientific applications, where reproducibility, traceability, and independent inspection are essential. By integrating natural-language interaction, domain Skills, computational execution, and structured reporting, GVIM 2.0 provides a practical framework for organizing computational chemistry and materials-informatics tasks. Several directions remain for further development. Larger external-validation studies, additional agent and tool-use baselines, more systematic error analysis, and prospective experimental case studies would further strengthen the evaluation. Future versions could also incorporate more specialized chemistry and materials Skills, broader database connectivity, and tighter integration with laboratory automation platforms. These extensions would help advance GVIM 2.0 from a workflow-level research-assistance system toward a more comprehensive scientific operating environment. Overall, GVIM 2.0 demonstrates how an agentic research system can translate natural-language scientific intent into executable, inspectable, and objectively scored workflows. Its main contribution is the provision of an integrated environment for organizing computational chemistry and materials-informatics tasks, rather than the replacement of domain experts or specialized scientific models. The results presented here support GVIM 2.0 as a reproducible research-assistance framework for chemistry and materials science and provide a foundation for the further development of more capable tool-augmented scientific AI systems.

## **Author contributions**

Kangyong Ma: Conceptualization, Methodology, Writing – original draft.

## **Conflicts of interest**

The authors declare no conflicts of interest.

## **Data availability**

The complete source code for this work is publicly available in our GitHub repository: <https://github.com/KangyongMa/GVIM-V2.0>.

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## References

- [1] Osaro, E.; Karpinski, N.; Aloroyo, S.; Ighalo, J. O. Artificial intelligence in materials science and chemistry: Past, present and future trajectories. *Materials Today Chemistry*, 2025, 49, 103115. DOI: 10.1016/j.mtchem.2025.103115.
- [2] Li, Y.; Wang, S.; Wang, J.; Qian, L.; Zhang, J. Artificial Intelligence for Materials Science: Transforming Research Paradigms. *Chemical Reviews*, 2026. DOI: 10.1021/acs.chemrev.6c00012.
- [3] Gupta, K. K.; Barman, S.; Sankar, A.; Dey, S.; Mukhopadhyay, T. Artificial Intelligence in Materials by Design: Critical Review and Perspectives on Materials Informatics to Generative and Agentic Intelligence. *Archives of Computational Methods in Engineering*, 2026, 33, 6015–6045. DOI: 10.1007/s11831-025-10486-3.
- [4] Mroz, A. M.; Basford, A. R.; Hastedt, F.; et al. Cross-disciplinary perspectives on the potential for artificial intelligence across chemistry. *Chemical Society Reviews*, 2025, 54, 5433–5469. DOI: 10.1039/d5cs00146c.
- [5] Ramos, M. C.; Collison, C. J.; White, A. D. A review of large language models and autonomous agents in chemistry. *Chemical Science*, 2025, 16, 2514–2572. DOI: 10.1039/d4sc03921a.
- [6] Vaswani, A.; Shazeer, N.; Parmar, N.; et al. Attention Is All You Need. *Advances in Neural Information Processing Systems*, 2017, 30.
- [7] OpenAI. Introducing ChatGPT. 2022.
- [8] Pei, Z.; Yin, J.; Zhang, J. Language models for materials discovery and sustainability: Progress, challenges, and opportunities. *Progress in Materials Science*, 2025, 154, 101495. DOI:

10.1016/j.pmatsci.2025.101495.

[9] Dagdelen, J.; Dunn, A.; Lee, S.; et al. Structured information extraction from scientific text with large language models. *Nature Communications*, 2024, 15, 1418. DOI: 10.1038/s41467-024-45563-x.

[10] Polak, M. P.; Schilter, O.; Le, T. Q.; et al. Flexible, model-agnostic method for materials data extraction from text using general purpose language models. *Digital Discovery*, 2024, 3, 1221–1235. DOI: 10.1039/d4dd00016a.

[11] Sumita, M.; Ishida, S.; Yoshizoe, K.; Tamura, R.; Terayama, K.; Tsuda, K. Molecular Design with Artificial Intelligence: Progress and Perspectives for Small Molecules. *Chemical Reviews*, 2026, 126, 3007–3054. DOI: 10.1021/acs.chemrev.5c00689.

[12] Jablonka, K. M.; Schwaller, P.; Ortega-Guerrero, A.; Smit, B. Leveraging large language models for predictive chemistry. *Nature Machine Intelligence*, 2024, 6, 161–169. DOI: 10.1038/s42256-023-00788-1.

[13] Guo, T.; Gu, Y.; Yang, X.; et al. What can large language models do in chemistry? A comprehensive benchmark on eight tasks. *Advances in Neural Information Processing Systems*, 2023, 36, 59662–59688.

[14] Lewis, P.; Perez, E.; Piktus, A.; et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*, 2020, 33, 9459–9474.

[15] Gao, Y.; Xiong, Y.; Gao, X.; et al. Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv:2312.10997*, 2023.

[16] Schick, T.; Dwivedi-Yu, J.; Dessì, R.; et al. Toolformer: Language Models Can Teach Themselves to Use Tools. *Advances in Neural Information Processing Systems*, 2023, 36.

[17] Yao, S.; Zhao, J.; Yu, D.; et al. ReAct: Synergizing Reasoning and Acting in Language Models. *International Conference on Learning Representations*, 2023.

[18] Wang, L.; Ma, C.; Feng, X.; et al. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 2024, 18, 186345. DOI: 10.1007/s11704-024-40231-1.

[19] Anthropic. Introducing the Model Context Protocol. 2024.

[20] Bran, A. M.; Cox, S.; Schilter, O.; Baldassari, C.; White, A. D.; Schwaller, P. Augmenting large language models with chemistry tools. *Nature Machine Intelligence*, 2024, 6, 525–535. DOI: 10.1038/s42256-024-

00832-8.

[21] Boiko, D. A.; MacKnight, R.; Kline, B.; Gomes, G. Autonomous chemical research with large language models. *Nature*, 2023, 624, 570–578. DOI: 10.1038/s41586-023-06792-0.

[22] Tom, G.; Schmid, S. P.; Baird, S. G.; et al. Self-Driving Laboratories for Chemistry and Materials Science. *Chemical Reviews*, 2024, 124, 9633–9732. DOI: 10.1021/acs.chemrev.4c00055.

[23] Häse, F.; Roch, L. M.; Aspuru-Guzik, A. Next-generation experimentation with self-driving laboratories. *Trends in Chemistry*, 2019, 1, 282–291. DOI: 10.1016/j.trechm.2019.02.007.

[24] Burger, B.; Maffettone, P. M.; Gusev, V. V.; et al. A mobile robotic chemist. *Nature*, 2020, 583, 237–241. DOI: 10.1038/s41586-020-2442-2.

[25] Kusne, A. G.; Yu, H.; Wu, C.; et al. On-the-fly closed-loop materials discovery via Bayesian active learning. *Nature Communications*, 2020, 11, 5966. DOI: 10.1038/s41467-020-19597-w.

[26] Raccuglia, P.; Elbert, K. C.; Adler, P. D. F.; et al. Machine-learning-assisted materials discovery using failed experiments. *Nature*, 2016, 533, 73–76. DOI: 10.1038/nature17439.

[27] Granda, J. M.; Donina, L.; Dragone, V.; Long, D.-L.; Cronin, L. Controlling an organic synthesis robot with machine learning to search for new reactivity. *Nature*, 2018, 559, 377–381. DOI: 10.1038/s41586-018-0307-8.

[28] MacLeod, B. P.; Parlane, F. G. L.; Morrissey, T. D.; et al. Self-driving laboratory for accelerated discovery of thin-film materials. *Science Advances*, 2020, 6, eaaz8867. DOI: 10.1126/sciadv.aaz8867.

[29] Shields, B. J.; Stevens, J.; Li, J.; et al. Bayesian reaction optimization as a tool for chemical synthesis. *Nature*, 2021, 590, 89–96. DOI: 10.1038/s41586-021-03213-y.

[30] Ji, Z.; Lee, N.; Frieske, R.; et al. Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 2023, 55, 1–38. DOI: 10.1145/3571730.

[31] Bender, E. M.; Gebru, T.; McMillan-Major, A.; Shmitchell, S. On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 2021, 610–623. DOI: 10.1145/3442188.3445922.

[32] Zhang, D.; Jia, X.; Tran, H. B.; Jang, S. H.; Zhang, L.; Sato, R.; Hashimoto, Y.; Sato, T.; Konno, K.;

Orimo, S.; Li, H. "DIVE" into hydrogen storage materials discovery with AI agents. *Chem. Sci.*, 2026, 17, 3031. DOI: 10.1039/D5SC09921H.

[33] Shen, Y.; Wang, L.; Huang, Y.; Zhang, X.; Huang, M.; Li, H.; He, J.; Cai, A.; Wang, Y.; Smith, P. E. S.; Jiang, J.; Zhu, Z.; Chen, L. Unlocking azobenzene isomerization mechanisms via an LLM agent-driven workflow integrating simulation, experiment, and machine learning. *Chem. Sci.*, 2026. DOI: 10.1039/D5SC08794E.

[34] Mirza, A.; Alampara, N.; Kunchapu, S.; et al. A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists. *Nat. Chem.*, 2025, 17, 1027-1034. DOI: 10.1038/s41557-025-01815-x.

[35] Zaki, M.; Krishnan, N. A. MaScQA: investigating materials science knowledge of large language models. *Digit. Discov.*, 2024, 3, 313-327. DOI: 10.1039/D3DD00188A.

[36] Wu, Z.; Ramsundar, B.; Feinberg, E. N.; et al. MoleculeNet: a benchmark for molecular machine learning. *Chem. Sci.*, 2018, 9, 513-530. DOI: 10.1039/C7SC02664A.

[37] Dunn, A.; Wang, Q.; Ganose, A.; Dopp, D.; Jain, A. Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm. *npj Comput. Mater.*, 2020, 6, 138. DOI: 10.1038/s41524-020-00406-3.

[38] Delaney, J. S. ESOL: estimating aqueous solubility directly from molecular structure. *J. Chem. Inf. Comput. Sci.*, 2004, 44, 1000-1005. DOI: 10.1021/ci034243x.

[39] Horai, H.; Arita, M.; Kanaya, S.; et al. MassBank: a public repository for sharing mass spectral data for life sciences. *J. Mass Spectrom.*, 2010, 45, 703-714. DOI: 10.1002/jms.1777.

[40] Nguyen, D. Q.; Zhai, Z.; Yoshikawa, H.; Fang, B.; Druckenbrodt, C.; Thorne, C.; Hoessel, R.; Akhondi, S. A.; Cohn, T.; Baldwin, T.; Verspoor, K. ChEMU: named entity recognition and event extraction of chemical reactions from patents. In *Advances in Information Retrieval, ECIR 2020*; Springer, 2020; pp 572-579. DOI: 10.1007/978-3-030-45442-5\_74.

[41] ByteDance, "DeerFlow: Deep Exploration and Efficient Research Flow," GitHub Repository. Available: <https://github.com/bytedance/deer-flow>

[42] Anthropic. Agent Skills. GitHub Repository. Available: <https://github.com/anthropics/skills>

